Simulating the Shared–AI–Vendor Effect

The idea in one paragraph

Many companies now use AI models (like GPT or Claude) to help make business decisions. Only a few companies ("vendors") supply these models, so many firms use the *same* model. If the model makes a mistake, all its users are hit by a similar mistake *at the same time*. Our hypothesis: firms using the same vendor should have stock returns that move together a little more than otherwise-similar firms. Your job is to build a small simulated stock market where we control the truth, and check that our statistical tests find the effect when it is there – and, just as important, find *nothing* when it is not.

Definitions

Vendor A supplier of AI models. We use two, called A and B.

Usage δ_i
The fraction of firm i's decisions made by its AI vendor. A number between 0 (no AI) and 0.4 (heavy use). Each firm has one vendor and one usage number.

Market factor m_t
A random number for each day t representing news that moves *all* stocks (the overall market).

Vendor shock $\eta_{A,t}$, $\eta_{B,t}$
A random number for each day representing how well or badly that vendor's model behaved that day. Crucially, it is the *same number for every firm using that vendor* on that day.

Firm noise ϵ_{it}
A random number specific to one firm on one day (its own good or bad luck). Different for every firm.

Theta θ
The strength of the vendor shock, between 0 and 1. $\theta = 0$ means AI errors are pure firm-specific luck (no common part). Bigger θ means more of the AI's mistakes are shared across its users. **This is the dial we turn.**

Residual return
What is left of a stock's return after removing the part explained by the market factor. We always compare firms using residuals, so that "everything moved because the market moved" is already taken out.

Same-vendor pair / cross-vendor pair
A pair of firms using the same vendor / different vendors.

Ground rules

Use any language you like (Python or R suggested). Set the random seed at the top of the script (`seed = 42`) so results reproduce exactly. Each part below says what to build, what to report, and what you should expect to see. If a check fails, stop and investigate before moving on.

1 Part 1: Build the fake market (half a day)

Create $N = 500$ firms and $T = 1{,}000$ trading days.

1. **Assign vendors.** Firms 1–300 use vendor A, firms 301–500 use vendor B (so A has a 60% share).

2. **Assign usage.** Draw each firm's δ_i from a uniform distribution on $[0, 0.4]$, once, and keep it fixed.

3. **Assign market sensitivity.** Draw each firm's β_i from a normal distribution with mean 1 and standard deviation 0.3, once.

4. **Draw the daily shocks.** For each day t: $m_t \sim N(0,1)$, $\eta_{A,t} \sim N(0,1)$, $\eta_{B,t} \sim N(0,1)$, and for each firm $\epsilon_{it} \sim N(0,1)$. All independent.

5. **Build returns.** For firm i on day t:

$$r_{it} = \underbrace{\beta_i m_t}_{\text{market part}} + \underbrace{\delta_i \sqrt{\theta}\, \eta_{v(i),t}}_{\text{shared AI part}} + \underbrace{\sqrt{1-\delta_i^2\theta}\ \epsilon_{it}}_{\text{firm's own luck}}$$

   where $v(i)$ is firm i's vendor. (The odd-looking last coefficient just keeps every firm's non-market variance equal to 1, so firms differ only in *how* they are risky, not in how risky they are. That is deliberate: it prevents the tests from picking up volatility differences instead of correlation.)

6. **Make two datasets.** Dataset ONE with $\theta = 0.2$ (effect exists). Dataset ZERO with $\theta = 0$ (no effect), using the *same* seed, vendors, usages, and betas – only the dial differs.

Report: a table with the number of firms, days, vendor shares, and the average δ_i; and one line confirming both datasets were built from identical firm assignments.

2 Part 2: Can we see the effect? (one day)

Work with residual returns throughout: regress each firm's r_{it} on m_t (one regression per firm, full sample) and keep the residuals.

1. **Pairwise correlations.** Draw 20,000 random pairs of firms. For each pair compute the correlation of their residuals over the 1,000 days, and record: same vendor or not, and the product $\delta_i \delta_j$.

2. **The headline comparison.** Compute the average correlation for same-vendor pairs and for cross-vendor pairs, in both datasets.

3. **The regression version.** Run

$$\text{corr}_{ij} = b_0 + b_1 \,\text{SameVendor}_{ij} + b_2 \,\text{SameVendor}_{ij} \times \delta_i\delta_j + b_3 \,\delta_i\delta_j + e_{ij}$$

in both datasets.

Report: Table – the 2×2 of average correlations (same/cross vendor × ONE/ZERO); Table – the regression coefficients with standard errors, both datasets side by side; Figure – binned scatter of pair correlation against $\delta_i\delta_j$, one line for same-vendor pairs, one for cross-vendor pairs, dataset ONE.

What you should see (check before proceeding): In dataset ONE: same-vendor correlations clearly above cross-vendor; b_2 positive and significant; b_3 near zero (two heavy AI users on *different* vendors are NOT extra-correlated – this is our key "placebo inside the design"). In dataset ZERO: everything near zero – same-vendor and cross-vendor look identical. If dataset ZERO shows an effect, something is wrong (most likely the market factor was not fully removed).

Why this matters: in real data, skeptics will say "same-vendor firms are just similar firms." Dataset ZERO shows our test does not manufacture an effect from shared assignment alone.

3 Part 3: Event days (one day)

Real AI vendors have identifiable "event days": new version releases and outages. In the simulation:

1. Pick 10 evenly spaced days as vendor A's **event days**. Rebuild dataset ONE identically except that on event days, vendor A's shock is drawn with five times the standard deviation: $\eta_{A,t} \sim N(0, 5^2)$.

2. For each day, compute the average correlation of vendor-A same-vendor pairs over a rolling 11-day window centered on that day (use the residuals; a simpler equivalent: the cross-sectional dispersion explained by the vendor-A average residual).

3. Also pick 10 *placebo days* (random non-event days) and repeat the comparison.

Report: Figure – the rolling same-vendor-A comovement measure over time with event days marked by vertical lines; Table – average comovement in event windows vs. placebo windows vs. all other days, for vendor-A pairs and (as a control) vendor-B pairs.

What you should see: visible spikes at the marked lines for vendor-A pairs only; placebo windows look like ordinary days; vendor-B pairs are flat throughout. This mimics how we will test the theory on real dates.

4 Part 4: A tiny equilibrium (one day, stretch goal)

So far usage δ_i was handed to firms. In the paper, firms *choose* it, facing a trade-off: AI improves decisions (a benefit) but exposure to a crowded vendor is penalized by the market (a charge that grows with how many others use the same vendor). Here is the whole thing with concrete numbers.

- Benefit of usage: $0.20 \times \delta$ for vendor A, $0.18 \times \delta$ for vendor B (vendor A is better).

- Cost of adopting: $\frac{1}{2}\delta^2$ (going all-in gets expensive).
- Crowding charge: $0.10 \times \delta \times D_v$, where D_v = (number of firms on vendor v ÷ total firms) × their average usage. D_v is the vendor's "crowd size."

Each firm picks the vendor and the usage δ that maximize (benefit − cost − charge), treating the crowd sizes as given. With the quadratic cost the best usage on vendor v has a formula: $\delta_v^* = \text{benefit rate}_v - 0.10 \times D_v$.

1. **Iterate to a fixed point.** Start from a 50/50 split with usage 0.1 everywhere. Loop: (i) compute each vendor's crowd size D_v; (ii) compute δ_v^* and each firm's achieved value on both vendors; (iii) move 5% of firms toward the vendor offering higher value; repeat until nothing changes (tolerance 10^{-6}).
2. **The experiment.** After convergence, improve vendor A: raise its benefit rate from 0.20 to 0.24 ("a better model ships"). Re-converge.
3. **Track two quantities** before and after: vendor A's share of firms, and total systematic variance $= 0.2 \times (D_A^2 + D_B^2)$ (using $\theta = 0.2$ from Part 1, this is the market-wide risk created by shared vendors).

Report: Table – share of firms on A, usage levels, crowd sizes, and total systematic variance, before vs. after the improvement; Figure – the convergence path of vendor A's share across iterations, both episodes on one plot.

What you should see: (1) Vendor A ends with more firms than B but *not all of them* – the crowding charge stops the better vendor from taking the whole market. (2) After the improvement, A's share and the total systematic variance both rise: **a better AI model makes the market as a whole riskier**, which is the punchline of the whole project. If you get 100% tipping to A, check the charge is applied per firm with the current D_v, not the starting one.

Deliverables summary

Part	Deliverable	Purpose
1	Setup table	Confirms the two datasets differ only in θ
2	2×2 correlation table	The effect exists (and only when it should)
2	Regression table (ONE vs. ZERO)	$b_2 > 0$, $b_3 \approx 0$; all ≈ 0 under ZERO
2	Binned scatter figure	Effect grows with joint usage
3	Rolling comovement figure	Spikes on event days only
3	Event vs. placebo table	Formal version of the spikes
4	Before/after equilibrium table	Better model ⇒ higher shared risk
4	Convergence figure	The market does not tip

Suggested pace: four working days plus writing. Send the Part 1–2 output before starting Parts 3–4. Code, config, and a two-page memo describing what you found (in words, not formulas) are the final deliverable.